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ATMOSPHERIC CHEMISTRY

Resetting tropospheric OH and CH₄ lifetime with ultraviolet H₂O absorptionMichael J. Prather^{1*} and Lei Zhu^{2,3†}

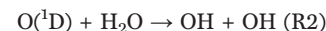
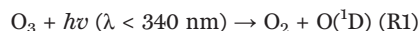
The decay of methyl chloroform, a banned ozone-depleting substance, has provided a clear observational metric of mean tropospheric hydroxyl radical (OH) abundance. Almost all current global chemistry models calculate about 15% too much OH and thus too rapid methane loss. Methane is a short-lived climate forcer, critical to achieving global warming targets, and this error affects our model projections of climate change. New observations of water vapor absorption in the ultraviolet region (290 to 350 nanometers) imply reductions in sunlight with key photolysis rates decreasing by 8 to 12% in the near-surface tropical atmosphere. Incorporation of this new mechanism in a chemistry-transport model reduces OH and methane loss by only 4%, but combined with other proposed mechanisms, such as tropospheric halogen chemistry (7%), we may be able to resolve this conundrum.

Projecting the efficacy of climate change mitigation efforts involving short-lived climate forcers such as the greenhouse gas methane (CH₄) (1) requires accurate modeling of its atmospheric loss, which is a function of its lifetime—the total atmospheric burden divided by the loss rate. A long-standing problem with most current global chemistry models (2, 3) is that the lifetime of CH₄ with respect to loss by tropospheric hydroxyl radical (OH), which constitutes about 82% of the total loss, is systematically lower than that scaled from the observed OH-driven decay of methyl chloroform (CH₃CCl₃) (4). We identified a missing component in current chemistry models—namely, ultraviolet (UV) absorption by water vapor (H₂O) (5)—and show that its inclusion in a photochemistry model will reduce OH levels, increasing the CH₄ lifetime, thus partly resolving the discrepancy found in most models (2).

A combination of recent laboratory studies, field measurements, and satellite observations presents a convincing case for absorption of radiation by H₂O at UV wavelengths (5, 6). Previously, only negligibly small UV H₂O absorption was found (7, 8). Both sides of that disagreement present plausible cases, and the difference remains unresolved. We took the recently measured H₂O cross sections, implemented them in a photolysis code, and examined the first-order impacts on tropospheric chemistry.

H₂O absorbs UV sunlight (290 to 350 nm) in the troposphere but does not photodissociate, and thus it reduces the overall photochemical activity. This tropospheric UV window coincides with that in which photolysis of ozone

(O₃) produces the metastable excited state of atomic oxygen, O(¹D) (reaction R1). Some of this O(¹D) reacts with H₂O (reaction R2), becoming the primary source of atmospheric hydroxyl radicals (OH). These OH radicals are responsible for the photochemical destruction of many pollutants, including CH₄ (reaction R3):



We adopted the measured UV spectrum for H₂O absorption from (5) as shown with the thin black line in Fig. 1A. This high-resolution (1 nm) absorption is mapped onto the broader wavelength bins used in the photolysis calculations of the Cloud-J module (Fig. 1A, blue bars) (9). These average H₂O cross sections are weighted by the variations in solar flux across each bin. Even these “large” H₂O cross sections are relatively small, only about eight times larger than the Rayleigh scattering cross section for air (Fig. 1A, black bars). The cross section for R1, scaled down by 10⁶, is shown in Fig. 1A as red bars. R1 occurs only for wavelengths of <340 nm (Fig. 1A, five leftmost bars), which have mid-point wavelengths of 295, 303, 310, 316, and 333 nm.

Photolysis results

We calculated the photolysis rate for R1, *J*_{OID} (per second), using Cloud-J 7.6 (10) updated to version 8.0 to include H₂O absorption in the UV region (Fig. 1A). Calculations in Fig. 1 are for overhead Sun, clear sky, and a tropical

Table 1. Reduction in key tropospheric photolysis rates (*J* values) at the surface when UV absorption by H₂O is included. Values were calculated for tropical atmosphere and overhead sun; lower sun angles increase the percent reduction. For all *J* values, reduction percent drops by ½ at ~3 km altitude and by ¼ at ~7 km. *J* values for long-lived (stratospheric) trace gases are not affected. Halogen species are not included. Other *J* values are reduced by <5%.

<i>J</i> value	Reduction at surface	Notes
O ₃ (1D)	11%	O ₃ photolysis yielding O(¹ D)
H ₂ COa	10%	H ₂ CO channel a yields H + HCO;
H ₂ COb	8%	channel b yields H ₂ + CO
H ₂ O ₂	9%	
CH ₃ OOH	8%	
N ₂ O ₅	8%	
HNO ₃	11%	
OCS	9%	
PAN	10%	
CH ₃ NO ₃	11%	
ActAld	12%	Acetaldehyde
MeVK	8%	Methylvinyl ketone
MeAcr	5%	Methacrolein
GlyAld	11%	Glycol aldehyde
MEKeto	11%	Methylethyl ketone
PrAld	11%	Propionaldehyde; C ₂ H ₅ CHO → C ₂ H ₅ +HCO
Glyxlb	9%	Glyoxal, (CHO) ₂ ; channel a
Glyxlc	9%	(HCO+HCO, 61% of total)
		is much less affected (3%) and not shown; channels b and c are equally affected
Acet-a	11%	Acetone; both channel a
Acet-b	12%	(CH ₃ CO+CH ₃) and channel b
		(CH ₃ +CH ₃ +CO) are affected almost equally

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atmosphere. The contribution to J_{O1D} from the five UV wavelength bins are shown in Fig. 1B. The 295-nm bin has the largest cross section for R1, but 99.5% of the photons in this bin are absorbed in the stratosphere, and so this bin contributes little to J_{O1D} (<5%). The 333-nm bin has very low cross sections for R1 and also contributes <5% to J_{O1D} . The dominant production of $\text{O}(^1\text{D})$ comes from the 303-, 310-, and 316-nm bins.

We show the total J_{O1D} tropospheric profile with (Fig. 1C, blue line) and without (Fig. 1C, red line) H_2O absorption. The decrease in J_{O1D} from previous models (without H_2O) is substantial: 11% near the surface, falling to 5% at 3 km altitude and 2% by 10 km. Many other key tropospheric photolysis rates also decrease at the surface (Table 1). For example, the photolysis rate for formaldehyde ($J_{\text{H}_2\text{CO}}$) is driven by UV wavelengths, and the two channels decrease by 8 to 10% (surface) to 4 to 5% (3 km) to 2% (10 km). By contrast, the photolysis rate for nitrogen dioxide (J_{NO_2}) with dominant cross sections at blue wavelengths decreases by only 1% throughout the troposphere. An obvious atmospheric test of UV H_2O absorption would be through profiles of measured and modeled actinic fluxes in the UV region as in (11), but absolute fluxes vary with nearby clouds and are difficult to model at the 10% level.

The UV-visible heating rates calculated in Cloud-J 7.6c consider only O_3 absorption and not H_2O absorption, and thus these rates are small, $<0.02 \text{ K day}^{-1}$ throughout most of the troposphere (Fig. 1D). With the H_2O UV absorption adopted here, these rates jump to 0.05 K day^{-1} near the surface, but this rate is inconsequential compared with that from solar heating by clouds [figure 3 in (12)], infrared absorption by H_2O , or other sources of available energy (13). In terms of the overall radiative balance in this case, the atmospheric absorption increases by 1.5 W m^{-2} (0.1%), reducing surface ocean absorption by 1.25 W m^{-2} and reflected sunlight by 0.25 W m^{-2} .

The photolysis code and underlying cross sections for other chemical species are described in (9) for Cloud-J version 7.3c and in (10) for Cloud-J and Solar-J version 7.6. The new photolysis module including absorption by H_2O (Cloud-J version 8.0 including its code, figures, and tables) is published in (14).

Chemistry results

The updated Cloud-J (v8.0) was incorporated into the current University of California, Irvine (UCI) chemistry-transport model (CTM). The UCI CTM, including Cloud-J, has been used for a variety of global chemistry studies (11, 15, 16), and in direct comparisons with many global models, the O_3 and OH chemistry results are consistent with those of the other top models (17–19). We made a 5-year simulation using the integrated forecast system (OpenIFS, cycle 38r1)

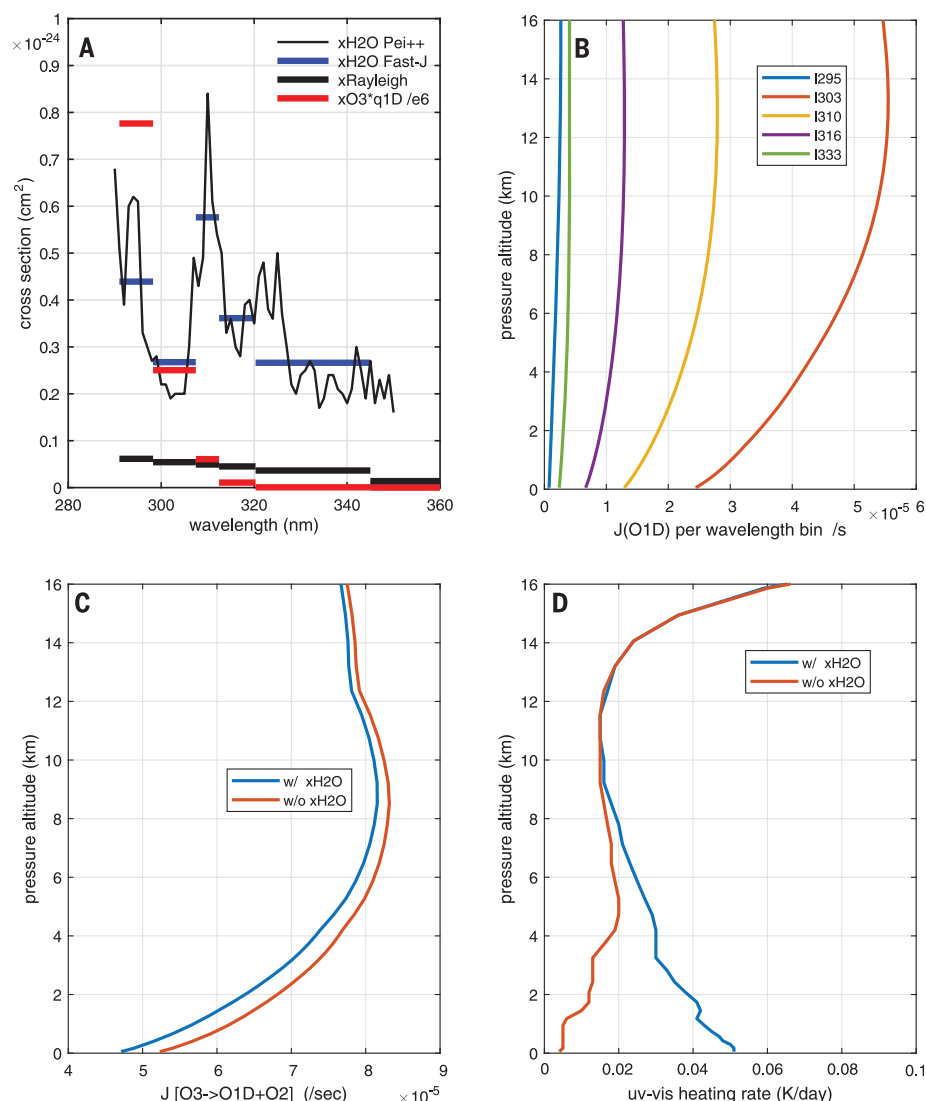


Fig. 1. UV absorption by H_2O and its impact on photolysis and heating rates. (A) UV cross sections for H_2O absorption from laboratory measurements (thin black line) (5), H_2O absorption as used by the wavelength bins in the Cloud-J (thick blue bars), Rayleigh scattering by air (thick black bars), and O_3 absorption times quantum yield producing of $\text{O}(^1\text{D})$ (scaled by 10^{-6} ; red bars). (B) Altitude profiles of J_{O1D} (per second) from each of the five UV wavelength bins in Cloud-J, centered on 295, 303, 310, 316, and 333 nm. Calculations were made with a tropical, cloud-free, overhead-sun, oceanic atmosphere, and total ozone column of 274 DU. Plotted are tropospheric values from surface (0 km, 1013 hPa, and 299.5 K) to tropopause (16 km, 110 hPa, and 191 K). The H_2O profile has a scale height of about 2.2 km and a column density of 14×10^{22} molecules cm^{-2} , which falls within the range of tropical atmospheres, 10×10^{22} to 18×10^{22} molecules cm^{-2} (27). (C) Altitude profiles of total J_{O1D} (per second) calculated with (blue) and without (red) H_2O UV absorption. (D) Altitude profiles of heating rates (kelvin per day) calculated by using only UV plus visible O_3 absorption (red) and then including UV H_2O absorption (blue).

meteorological data for years 2000 to 2004 but with annually repeating emissions for year 2000 from scenario RCP-6.0. The annual chemical budgets averaged over the last 4 years for O_3 , carbon monoxide (CO), and CH_4 are given in Table 2. We did not activate the extra diagnostics for stratosphere-troposphere exchange of O_3 in these CTM simulations, and so it was derived from mass balance.

The budgets in teramoles provide for some notable comparisons. About 60% of the CO source is from in situ chemical production ($64 \text{ Tmol year}^{-1}$), about half of which is from CH_4 oxidation ($34 \text{ Tmol year}^{-1}$), and the rest is from short-lived nonmethane hydrocarbons. The primary OH production ($101 \text{ Tmol year}^{-1}$) matches the OH oxidation of CO ($99 \text{ Tmol year}^{-1}$), and thus the extra OH to oxidize CH_4 and other

Table 2. Reference case tropospheric chemical budgets (teramole, teramole per year, and year and change (%)). Chemical production and loss refer to tropospheric chemistry only; stratospheric chemistry is counted as stratosphere-troposphere flux (teramoles per year). Results are the average of the last 4 years (2001–2004) of a UCI CTM simulation by using constant year 2000 emissions from the RCP6.0 scenario. The terms were taken from the budget tendencies except for the O ₃ s-t flux, which was derived from mass balance. The O ₃ production and loss terms assume that ground-state atomic oxygen O(³ P) and O ₃ are equivalent, and thus rates such as O+O ₂ +M→O ₃ are not included, but the rate NO ₂ +hν→NO+O counts as the dominant O ₃ +O production (>99%) in these budgets. The rates O(¹ D)+H ₂ O (50 Tmol year ^{−1}) and O(¹ D)+CH ₄ (0.02 Tmol year ^{−1}) count as O ₃ +O loss, whereas O ₂ +hν counts as production of two O ₃ +O (0.6 Tmol year ^{−1}). CH ₄ and CO chemical loss in these simulations is 99.9% through tropospheric OH. Changes in radicals other than OH were smaller: NO and NO ₂ tropospheric mass changed by <0.1%; HO ₂ decreased by 0.9%. The lifetime versus OH is a simple burden:loss ratio; no chemical feedbacks are included.				
	Tropospheric O ₃ + O	Global CO	Global CH ₄	OH through O(¹ D)+H ₂ O
Mass (Tmol)	7.2	11.9	306.4	
Emission (Tmol year ^{−1})	None	+45.54	lower boundary condition	
Chemical production (Tmol year ^{−1})	+507.34	+64.05	None	100.6
Chemical loss (Tmol year ^{−1})	−501.64	−99.42	−33.73	
Surface deposition (Tmol year ^{−1})	−13.24	−4.77	−1.88	
Stratosphere-troposphere flux (Tmol year ^{−1})	+7.54	−5.38	−1.99	
Net sum (Tmol year ^{−1})	0.00	0.02	37.60	
Lifetime versus OH (years)		0.120	9.08	
Relative change (%) in magnitude caused by UV H ₂ O absorption				
Mass	+2.05%	+2.09%	+0.02%	
Chemical production	+0.90%	−2.48%	none	−4.42%
Chemical loss	+0.85%	−1.77%	−4.09%	

hydrocarbons must be part of OH amplification involving reactions of OH, hydroperoxyl radical (HO₂), nitric oxide (NO), nitrogen dioxide (NO₂), and hydrocarbons.

A parallel perturbation simulation was run with the H₂O absorption cross sections zeroed out, and the percent changes are given in Table 2. Because the H₂O absorption results in 8 to 12% boundary-layer reductions in photolysis rates for so many species [such as formaldehyde (H₂CO), hydrogen peroxide (HOOH), acetic acid (CH₃OOH), acetaldehyde, and acetone (Table 1)], we expected complex changes across the major tropospheric species. The largest perturbation caused by the H₂O absorption was still what we expected from the photolysis-only example above: Primary OH production is reduced by 4.4%, and the OH-driven loss of CH₄ decreases by 4.1%. This decrease results in a reduced CH₄ source of CO, but the CO sink is also reduced by 3.8%, so that overall CO increases by 2.1%. The OH reduction has a larger impact on CH₄ compared with that on CO (4.1 versus 3.8%) because the CH₄ loss is highly temperature dependent, occurring more rapidly in the lowermost tropical

troposphere where the H₂O absorption has the largest reduction in J_{OH} (Fig. 1).

Tropospheric O₃ increases by 2.0%, which is hard to explain simply with the budget terms we have available. The obvious explanation is that O₃ loss was reduced by 2 to 4% because the major terms are O(¹D)+H₂O, O₃+OH, and O₃+HO₂. The first two terms dropped 4%, and HO₂ dropped only 1%. We assigned production of O₃ to the rates of HO₂+NO and other peroxy radicals with NO. The HO₂ mass dropped only 1%, whereas NO and NO₂ were unchanged. Thus, the photolysis changes reduced O₃ loss more than production, and the abundance increased.

The lower tropical troposphere dominates both the CH₄ budget and the impact of H₂O absorption. The tropospheric loss of CH₄ to reaction with OH is weighted toward the lower troposphere, with 40% of the total occurring in the 800 to 1000 hPa range, 84% in the 500 to 1000 hPa range, and about 8% at pressures <400 hPa. The relative reduction in CH₄ loss from H₂O absorption is about 2.5% at pressures of <500 hPa, increasing linearly in pressure to 6.5% at 1000 hPa. Half (50%) of

CH₄ loss occurs in the core tropics (±20° latitude), where the impact of H₂O absorption is largest (>5%), and most loss (90%) occurs within ±45° latitude, where the H₂O impact drops to about 3%. The longitudinal impact of H₂O varies about 4.3%, with a standard deviation of 0.6%. Thus, the largest relative impact of UV absorption by H₂O occurs where the absolute loss of CH₄ is greatest.

Discussion

This process—UV absorption by H₂O—can only partially resolve one of the more enduring conundrums in modeling CH₄: Use of the OH calculated in current models produces a much more rapid decay of atmospheric CH₃CCl₃ than is observed, 0.18 year^{−1} (±1%) over 1998 to 2008 (16, 20). From table 1 of (2), the mean modeled CH₄ lifetime across 16 global chemistry models is biased 15% low (biased high in OH). Work by (3) examined 10 next-generation models and found that their CH₄ lifetime for the 2000 period ranged from 6.6 to 8.5 years, with most having lifetimes below 8 years, which also falls well below an observationally based range of 9.1 ± 0.9 years (16). Nicely *et al.* (3) also identified the tropospheric O₃ abundances and J_{OH} as the primary factors that control model differences. Unfortunately, we can get only 4% of the 15% OH reduction by including H₂O absorption and thus must continue to look elsewhere.

A number of chemistry models have added tropospheric halogen chemistry (21–24) and found that the ensuing chemical changes typically caused an 8% reduction in the OH+CH₄ rate, which was attributed to a 20% reduction in present-day tropospheric O₃—for example, from 30 to 24 Dobson units (DU). Calculations made using the NASA Atmospheric Tomography mission (ATom) parcel data (25) show that O₃ controls CH₄ loss with a sensitivity factor of 0.46 over the Atlantic and Pacific Ocean basins, and combining this with a 20% reduction in tropospheric O₃ gives a 9% reduction in OH+CH₄. One challenge for the halogen model is that the major shift in tropospheric O₃ down to 24 DU may be incompatible with observations that give values closer to 30 DU (Table 1) (26).

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